Lab 07 – Contribution – Alexander DeBusk

- Reformatted the Python code in the notebook so each statement was on its own line and easier to read.
- Helped complete the model training section using Logistic Regression to predict Titanic passenger survival.
- Added the confusion matrix and heatmap to visualize correct and incorrect predictions.
- Used the classification report to examine precision, recall, and F1-scores.
- Completed the five-fold cross-validation section and included the average score and standard deviation.
- Answered the knowledge check questions about precision, recall, and cross-validation.
- Reviewed the purpose of each evaluation technique and how they can be used in real-world situations.
- Prepared the notebook content and helped organize the required PDF deliverables.
- Submitted my work on time all with correct naming conventions for all deliverables.

CONTRIBUTION JOURNAL – Module 07: Model Evaluation

ITAI 1371

Student: Shareefah Adeola

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Overview of My Contribution - In Module 07, I worked on evaluating a machine learning model using the Titanic dataset. I completed the coding tasks in Google Colab, reviewed the outputs, interpreted the model evaluation metrics, and answered the Knowledge Check questions.

Model Setup and Training - I imported the required Python libraries, loaded the Titanic dataset, selected Age, Passenger Class, Sex, and Fare as features, and used Survived as the target variable. I replaced missing age values with the median and encoded the Sex variable numerically, splited the data into training and testing sets and trained a Logistic Regression model. The model trained successfully using 712 training rows and 179 testing rows.

Confusion Matrix - I used the confusion_matrix function from scikit-learn and visualized the results using Seaborn's heatmap. The matrix showed 144 correct predictions and 35 incorrect predictions. I interpreted the four outcomes to understand where the model predicted correctly and where it made errors.

Classification Report - I generated a classification report using classification_report from scikit-learn. The report provided precision, recall, F1-score, and support for both classes. The overall accuracy was approximately 80%, and I reviewed the differences between the model's performance for passengers who survived and those who did not survive.

Cross-Validation - I used cross_val_score to perform five-fold cross-validation. The average score was 0.7868, or approximately 78.68%, and the standard deviation was 0.0094. I reviewed the

scores from the five folds and learned how cross-validation helps assess the consistency of model performance.

Knowledge Check - I completed the three Knowledge Check questions covering situations where precision is more important than recall, situations where recall is more important than precision, and why cross-validation is generally more reliable than a single train-test split.

Summary - My contribution to this module included writing and running the required code, troubleshooting a Python import error, generating the model evaluation outputs, interpreting the results, and completing the Knowledge Check. Through these activities, I strengthened my practical understanding of classification metrics and model evaluation in Python.

Contribution

In this lab, I completed the main coding tasks for evaluating a machine learning model using the Titanic dataset. I worked with different evaluation methods to better understand the model's performance and learn how to identify its mistakes. I used the confusion matrix to compare the model's predictions with the actual results. I also generated a classification report to check precision, recall, and F1 scores. These results helped me understand the types of mistakes the model made and why accuracy alone is not enough to evaluate a model. I used 5-fold cross validation to evaluate the logistic regression model on different parts of the dataset. I calculated the scores for each fold, the average score, and the standard deviation. This helped me understand how cross validation can provide a more reliable view of model performance.

I reviewed the evaluation methods and answered the knowledge to check questions about precision, recall, and cross validation. I also explained how different evaluation metrics can be used in real world situations. Through this lab, I improved my ability to evaluate machine learning models and understand their strengths and limitations. These skills will be useful in my future machine learning projects.